Shivam Kumar

Aspiring AI/ML Engineer | **Generative AI & NLP** | **Machine Learning**

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Projects

RAG-Based AI Teaching Assistant | GitHub — Jan. 2026 – Jul. 2026

- Architected an end-to-end **Retrieval-Augmented Generation (RAG)** pipeline in **Python** using embeddings, **FAISS** vector search, **Hugging Face Transformers**, and **LLMs** for context-aware academic question answering.
- Integrated **Whisper** speech-to-text with semantic retrieval and LLM response generation, **reducing irrelevant responses by 30%** through optimized retrieval logic and prompt engineering.
- Deployed a production-ready **Gradio** application on **Hugging Face Spaces**, enabling real-time text and voice-based Q&A for end users.

Abhedya Chakra – Mule Account Risk Screening | GitHub — Oct. 2026

- Built a full-stack financial transaction investigation prototype using **FastAPI** and **React** to support account-level risk screening and transaction analysis.
- Implemented risk-signal analysis and **Isolation Forest** anomaly detection to flag unusual account behavior for further investigation.
- Added transaction-network visualization, candidate fund-tracing paths, account investigation details, and downloadable screening reports.

ClassMate AI – Face Recognition Attendance System | GitHub | Live Demo — Oct. 2026

- Developed a **Python**-based classroom management application with a **Streamlit** interface and dedicated teacher and student portals.
- Integrated a face-recognition attendance workflow and connected application data services using **Supabase**, with **scikit-learn** and **dlib**-based dependencies.
- Deployed the web app on **Streamlit Community Cloud** and maintained the source code and dependency setup on GitHub.

Technical Skills

Languages & Core CS: Python, SQL, Data Structures & Algorithms, Object-Oriented Programming (OOP), Git, GitHub

Machine Learning: Supervised & Unsupervised Learning, Regression, Classification, Clustering, Feature Engineering, Model Training & Evaluation, Cross-Validation, Hyperparameter Tuning, Scikit-learn

Deep Learning & NLP: Neural Networks, PyTorch, TensorFlow, Transformers, Transfer Learning, Model Fine-Tuning, NLP, Whisper, Text Embeddings

Generative AI & LLM Engineering: LLMs, RAG (Retrieval-Augmented Generation), Prompt Engineering, LangChain, FAISS, Vector Databases, LLM APIs, AI Agents, Agentic AI, MCP

Data & Backend: NumPy, Pandas, Matplotlib, Seaborn, Flask, FastAPI, Hugging Face Transformers, REST APIs, PostgreSQL, SQLite

MLOps & Deployment: Hugging Face Spaces, AWS, GitHub Actions, Model & Cloud Deployment

Education

Sage University — Jun. 2024 – Jun. 2028
B.Tech. in Artificial Intelligence & Machine Learning | *CGPA: 7.57/10* — *Indore, India*

Certifications

- **Python Programming with Red Hat (AD141-RHA), Ver. 9.0** — Dec. 2025
- **PMR Certification – SAMTRIX** — Dec. 2025
- **AWS Academy Graduate – Machine Learning Foundations** — Dec. 2025

Experience

- Participated in **VOID HACKS**, gaining hands-on experience in collaborative problem-solving and rapid software development. LinkedIn Post.